

The empirical recurrence for OEIS A300339, recovered from its tail

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Abstract

OEIS A300339 records “Empirical recurrence of order 66” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 111$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 70$. Its last coefficient is $c_{66} = 33169152 \neq 0$, so no further tail satisfies anything shorter; any order-66 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A300339 is “Number of $n \times 5$ 0..1 arrays with every element equal to 0, 2, 3, 4, 5, 6, 7 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

1, 127, 1272, 25327, 476507, 9482056, 193080853, 3949300187, ...

The entry states:

Empirical recurrence of order 66 (see link above)

As of the “Last modified” line on the live entry (Mar 03 2018 08:17:15, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 111$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 111$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 66: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 66.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 222$ exact terms from $n = 4$ on, it returns order 66 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{66} c_i a(n-i),$$

c_1	23	c_{18}	−1264242485	c_{35}	14858818624233	c_{52}	−3087415787040
c_2	60	c_{19}	−78751885200	c_{36}	−2779226378227	c_{53}	−910038796806
c_3	−2106	c_{20}	46014187029	c_{37}	−10140418555217	c_{54}	1547020047646
c_4	−7557	c_{21}	228066822233	c_{38}	−5591985101384	c_{55}	618772080692
c_5	90253	c_{22}	−316341047121	c_{39}	11277180527122	c_{56}	−377905458158
c_6	255514	c_{23}	−292207835108	c_{40}	7964372885520	c_{57}	−167072983588
c_7	−1540418	c_{24}	1061312250118	c_{41}	2981850813510	c_{58}	13803271442
c_8	−4629304	c_{25}	−433008559353	c_{42}	−9342307273259	c_{59}	33799838976
c_9	16629869	c_{26}	−2168295704375	c_{43}	−31351651154855	c_{60}	−1271568412
c_{10}	41052020	c_{27}	2453222255253	c_{44}	20413313327775	c_{61}	−12079714576
c_{11}	−135122565	c_{28}	2287670802706	c_{45}	36539358392936	c_{62}	3933207856
c_{12}	−220492294	c_{29}	−6077196812589	c_{46}	−16588953394581	c_{63}	2310948512
c_{13}	980299723	c_{30}	1598446292598	c_{47}	−20604513837289	c_{64}	−686981344
c_{14}	707253160	c_{31}	12527689438467	c_{48}	6261624481691	c_{65}	−118987584
c_{15}	−5380205879	c_{32}	−8630016151796	c_{49}	3507738971612	c_{66}	33169152
c_{16}	−1063503128	c_{33}	−18138899569708	c_{50}	1265045783745		
c_{17}	21372870701	c_{34}	9056993743844	c_{51}	1010138582211		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 70$, and it is the recurrence OEIS A300339 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{66} - c_1 x^{65} - \dots - c_{66}$. Its constant term is $-c_{66} = -33169152$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with

$\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 66 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 66, so $\deg q = 66$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 66, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 66, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 33169152.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-66 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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